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Course: Advanced Networking and Data Security

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Date: 13 April 2026

Introduction

In today's digital world, understanding how networks work is crucial. In my view, tools like Wireshark play a pivotal role in helping us analyze and understand network communications in a practical way. Wireshark is a network packet analyzer that captures and displays data packets in great detail, allowing users to see what's happening within the network in real time.

What makes Wireshark particularly useful is that it provides in-depth insight into how protocols like HTTP, TCP, and IP operate in real time. This makes it valuable not only for learning purposes but also for network troubleshooting and security improvement. In the past, tools like these were expensive and limited in their capabilities, but Wireshark has made network analysis much more accessible thanks to its open-source nature (Wireshark documentation, 2026).

Overall, I consider Wireshark an essential tool for students, network administrators, and security professionals who want to better understand, manage, and secure networks.

System Specifications Section:

Before installing Wireshark, it was necessary to verify the system specifications to ensure compatibility. My laptop, which I used for this task, was running Windows 10 Pro 64-bit (version 22H2, build number 19045.7058) with an Intel Core i5-7200U processor clocked at 2.50 GHz. It had 8.00 GB of RAM, a 238 GB Samsung SSD, and an Intel HD Graphics 620 card. According to the Wireshark user manual, it is important to check system compatibility before installation to ensure the tool functions correctly (Sharpe et al., n.d.). Based on these specifications, my system met all the requirements for installing and running Wireshark version 4.6.4 without any issues.

Wireshark Environment Section:

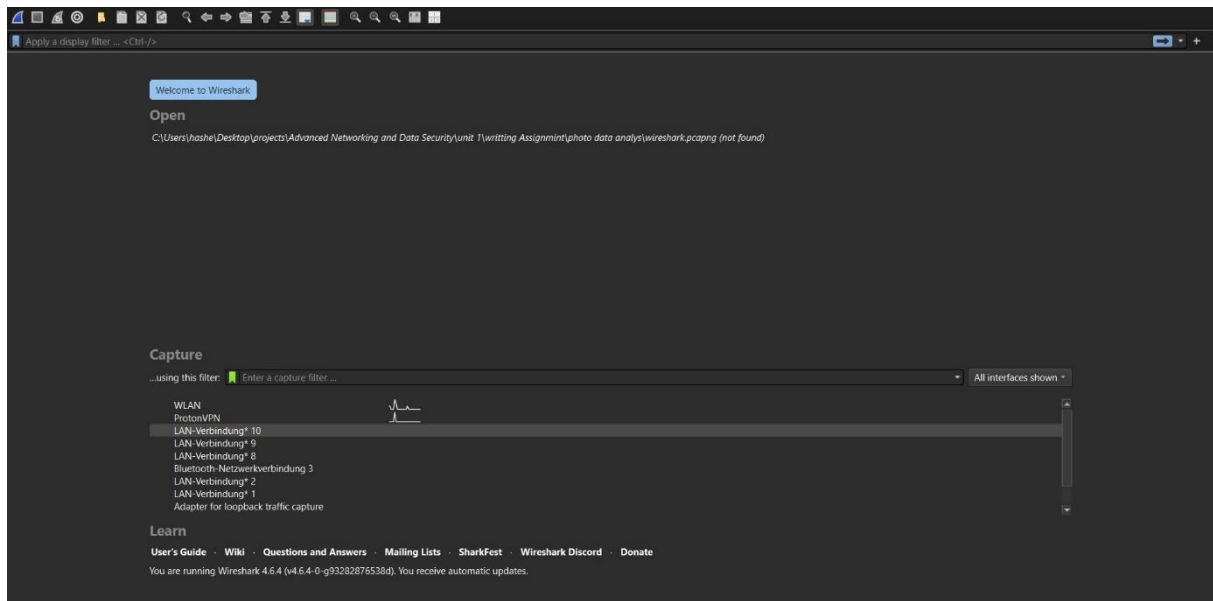
Upon launching Wireshark, the interface immediately displayed several network interfaces, each with a small activity graph next to it. This made it easy to identify which interfaces were actively transmitting data. For example, the WLAN interface showed clear activity, indicating that the Wi-Fi connection was in use, while the ProtonVPN interface also showed some movement due to the installed VPN software. This visual representation helped me quickly understand which connections were active on my device. When I double-clicked on a captured packet to expand its details, I was fascinated by the layers of information displayed, including the Ethernet frame, IP address data, and protocol details. The raw data looked exactly like the kind of screens you see in hacker movies, but now I understand that this is simply how real network traffic looks when analyzed at a technical level. According to the Wireshark User's Guide, each packet contains multiple layers of information that correspond to the different layers of the OSI model, which helps network professionals analyze and troubleshoot communication issues (Sharpe et al., n.d.).

1)

When I first opened the Wireshark interface, I noticed several network interfaces (gateways). These included WLAN, which represents my Wi-Fi connection, ProtonVPN, which appears because I have a VPN installed and use ProtonVPN, multiple LAN-Verbindung entries indicating wired connections, and Ethernet, which refers to the network cable connection.

After that, I closed Google Chrome, including all active browser tabs, and then launched the Wireshark application. At that moment, no data packets were displayed because Wireshark had not yet captured any network activity.

format 1 :showing no network activity detected



2)

After logging into the University of the People's student portal, I navigated to both the announcements section and the course forum. Then I returned to Wireshark and selected the wireless (WLAN) interface since I was connected to Wi-Fi. Immediately, a large amount of data appeared on the screen. At first, the information seemed overwhelming and confusing, but this is perfectly normal for a new user. According to the Wireshark user manual, this data represents all network packets sent and received during the browsing session, including details such as the source address, destination address, protocol used, and the size of each packet.

format 2 :The data packet capture

The image shows a Wireshark interface with a packet list on the left and a packet details pane on the right. The packet list contains 13 packets, all of which are Transport Data packets from 146.70.246.98 to 10.239.235.222. The packet details pane shows the selected packet (No. 13) with its raw data and a hex dump.

No.	Time	Source	Destination	Protocol	Length	Info
8543	205.808232900	146.70.246.98	10.239.235.222	WireGu...	122	Transport Data, receiver=0x8D602D67, counter=117, datalen=48
8544	205.808232900	146.70.246.98	10.239.235.222	WireGu...	122	Transport Data, receiver=0x8D602D67, counter=118, datalen=48
8545	205.808232900	146.70.246.98	10.239.235.222	WireGu...	122	Transport Data, receiver=0x8D602D67, counter=119, datalen=48
8546	205.808232900	146.70.246.98	10.239.235.222	WireGu...	122	Transport Data, receiver=0x8D602D67, counter=120, datalen=48
8547	205.808232900	146.70.246.98	10.239.235.222	WireGu...	122	Transport Data, receiver=0x8D602D67, counter=121, datalen=48
8548	205.832184700	146.70.246.98	10.239.235.222	WireGu...	154	Transport Data, receiver=0x8D602D67, counter=122, datalen=80
8549	205.832184700	146.70.246.98	10.239.235.222	WireGu...	154	Transport Data, receiver=0x8D602D67, counter=123, datalen=80
8550	205.801880000	10.239.235.222	146.70.246.98	WireGu...	122	Transport Data, receiver=0x7E751368, counter=113, datalen=48
8551	205.882247200	10.239.235.222	146.70.246.98	WireGu...	122	Transport Data, receiver=0x7E751368, counter=114, datalen=48
8552	205.928962500	146.70.246.98	10.239.235.222	WireGu...	266	Transport Data, receiver=0x8D602D67, counter=124, datalen=192
8553	205.928962500	146.70.246.98	10.239.235.222	WireGu...	266	Transport Data, receiver=0x8D602D67, counter=125, datalen=192
8554	205.930919200	10.239.235.222	146.70.246.98	WireGu...	154	Transport Data, receiver=0x7E751368, counter=115, datalen=80
8555	205.931105000	10.239.235.222	146.70.246.98	WireGu...	154	Transport Data, receiver=0x7E751368, counter=116, datalen=80
8556	205.931201500	10.239.235.222	146.70.246.98	WireGu...	154	Transport Data, receiver=0x7E751368, counter=117, datalen=80
8557	205.931269300	10.239.235.222	146.70.246.98	WireGu...	154	Transport Data, receiver=0x7E751368, counter=118, datalen=80
8558	205.973732600	146.70.246.98	10.239.235.222	WireGu...	122	Transport Data, receiver=0x8D602D67, counter=126, datalen=48
8559	205.978656000	146.70.246.98	10.239.235.222	WireGu...	122	Transport Data, receiver=0x8D602D67, counter=127, datalen=48
8560	206.504845700	146.70.246.98	10.239.235.222	WireGu...	138	Transport Data, receiver=0x8D602D67, counter=128, datalen=64
8561	206.504845700	146.70.246.98	10.239.235.222	WireGu...	122	Transport Data, receiver=0x8D602D67, counter=129, datalen=48
8562	206.505005400	10.239.235.222	146.70.246.98	WireGu...	122	Transport Data, receiver=0x7E751368, counter=119, datalen=48
8563	206.505198100	10.239.235.222	146.70.246.98	WireGu...	122	Transport Data, receiver=0x7E751368, counter=120, datalen=48
8564	206.557147100	146.70.246.98	10.239.235.222	WireGu...	122	Transport Data, receiver=0x8D602D67, counter=130, datalen=48
8565	206.868645500	146.70.246.98	10.239.235.222	WireGu...	138	Transport Data, receiver=0x8D602D67, counter=131, datalen=64

Packet 13 details:

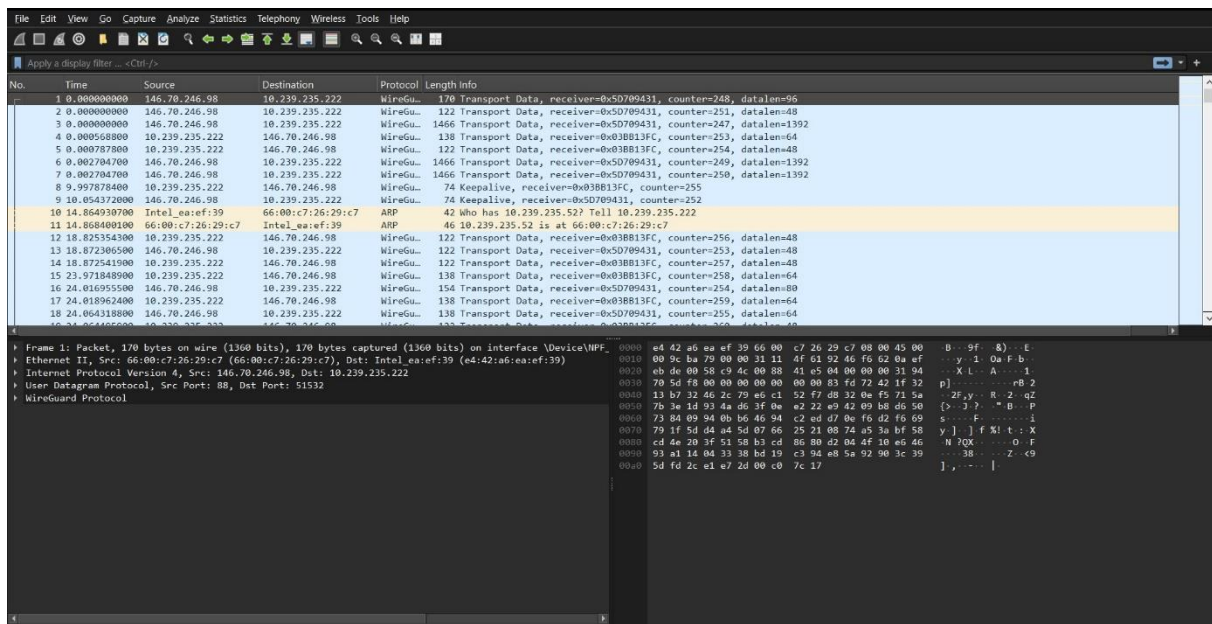
- Frame 13: Packet, 170 bytes on wire (1360 bits), 170 bytes captured (1360 bits) on interface \Device\NPF...
- Ethernet II, Src: 66:00:c7:26:29:c7, Dst: Intel ea:ef:35 (e4:42:a6:ea:ef:35)
- Internet Protocol Version 4, Src: 146.70.246.98, Dst: 10.239.235.222
- User Datagram Protocol, Src Port: 88, Dst Port: 51532
- WireGuard Protocol

Raw data (hex): 0000 e4 42 a6 ea ef 35 66 00 c7 26 29 c7 00 00 45 00 00 00 9c ba 79 00 00 31 11 4f 61 92 46 f6 62 0a ef 00 00 eb de 00 58 c9 4c 00 88 41 e5 04 00 00 00 31 94 00 3b 70 5d f8 00 00 00 00 00 00 83 fd 72 42 1f 32 00 00 13 b7 32 45 2c 79 e6 c1 52 f7 00 32 0e f5 71 5a 00 5d 7b 3c 1d 93 4a d5 3f 0a e2 22 a9 42 09 b8 d5 50 00 6d 73 84 09 94 0b b6 46 94 c2 ed d7 0e f6 d2 f6 69 00 7b 79 1f 5d 4a a4 5d 07 65 25 21 08 74 a5 3a bf 58 00 00 cd 4e 20 3f 51 5b b3 c1 86 80 d2 04 4f 10 e6 46 00 00 93 a1 14 04 33 38 bd 19 c3 94 a8 5a 92 90 3c 39 00 a0 5d fd 2c e1 e7 2d 00 c8 7c 17

3)

After approximately 30 seconds, I stopped the capture by navigating to the Capture menu and clicking Stop. I then saved the file using the File menu. This hands-on experience taught me several things. First, I learned how to capture network traffic and identify the correct network interface. Second, I became familiar with the Wireshark environment, which I had always wanted to explore but never had the chance to. Before this assignment, Wireshark seemed like a very complex and intimidating tool. However, after actually using it, I realized that it is much more approachable than I expected. This practical experience helped me break that mental barrier and gave me the confidence to explore it further in future units (Sharpe et al., n.d.).

Format 3 : Saving data after capture

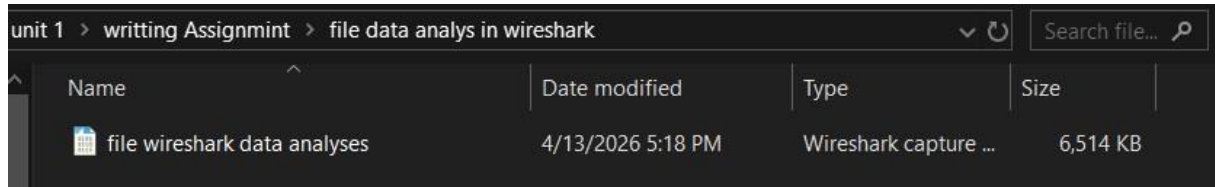


Conclusion:

In conclusion, this assignment was a valuable and practical introduction to network analysis. Installing and using Wireshark for the first time gave me real hands-on experience that goes beyond theoretical knowledge. As someone who aspires to become a security engineer and work in the cybersecurity field, I found this experience especially meaningful. Wireshark gave me my first real entry point into the practical side of network security, and I look forward to developing these skills further in the coming units (Sharpe et al., n.d.).

❖ Additions :

Image of the saved file location



The image shows a file explorer window with a dark theme. The breadcrumb path at the top is 'unit 1 > writting Assignmint > file data analys in wireshark'. There is a search bar on the right labeled 'Search file...' with a magnifying glass icon. Below the breadcrumb, there is a table with four columns: 'Name', 'Date modified', 'Type', and 'Size'. A single file is listed in the table.

Name	Date modified	Type	Size
 file wireshark data analyses	4/13/2026 5:18 PM	Wireshark capture ...	6,514 KB

References:

- Sharpe, R., Warnicke, E., & Lamping, U. (n.d.). *Wireshark user's guide*. Wireshark. https://www.wireshark.org/docs/wsug_html_chunked/index.html
- Wireshark Documentation. (2026). *Wireshark*. <https://www.wireshark.org/docs/>
- NetworkTutor. (2022, October 1). Wireshark - Beginners guide - 101 | How to install and capture packets | How to filter ICMP | TCP [Video]. YouTube. <https://www.youtube.com/watch?v=Ud0QK0TPu4U>